Homework 1 Cover Page.

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Date of Submission: 02/03/2026

Problem set Number: #1

Due Date: 02/03/2026

Acknowledgement: No help.

Problem 1.

The most dominant heat transfer mode in outer space is **radiation**. As the outer space is vacuum, there's no gas or liquid for convection to occur, plus the fact that the spacecraft is not in contact with any physical objects makes conduction not possible as well. Specifically, the spacecraft would absorb the solar radiation from the sun and Earth's albedo, and reject the heat by rejecting or reflecting the radiation.

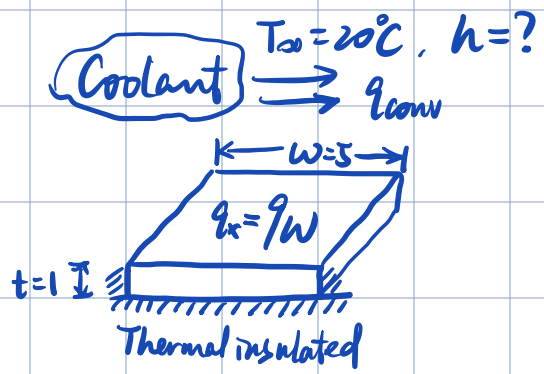
The most dominant heat transfer mode at the atmosphere would be **convection**. As the spacecraft travels through the atmosphere, the extremely high speed would compress and heat the gas surrounding it, and form a heated layer around the spacecraft. In this case, convective heat transfer would occur between the spacecraft and the hot accelerated surrounding gas.

Problem 2.

Known: heat transfer rate ($q_x = 9W$), size of chip ($w = 5mm$, $t = 1mm$), thermal conductivity ($k = 150 W/mK$), Coolant temperature ($T_\infty = 20°C$)

Find: (a) Temperature difference (b) minimum heat transfer coefficient; what convection process to use.

Schematic:



$$q_x = -kA\frac{dT}{dx}$$

Assumptions: (1) Thermal radiation is negligible. (2) Bottom and side are insulated thus no heat transfer (3) steady state

Analysis: (a) For the top surface, convection occurs between the chip and the coolant, where

$$q_x = -kA\frac{dT}{dx}, \text{ where } A = w^2 = 25\,mm^2$$

$$\frac{dT}{dx} = \frac{-q_x}{kA} = \frac{-9W}{150\,W/mK \cdot 25\cdot 10^{-6}\,m^2} = -2400\,\frac{K}{m}$$

$$T = -2400\,x + T_{front}$$

$$T_{back} = -2400\,\frac{K}{m} \cdot (1\cdot 10^{-3}\,m) + T_{front} = 2.4\,K + T_{front}$$

$$\therefore \Delta T = T_{back} - T_{front} = \boxed{2.4\,K}$$

(b) Looking at the convective flow, the heat transfer rate is:

$$q = hA(T_s - T_\infty)$$

Thus rearrange to get the heat transfer coefficient h:

$$h = \frac{q}{A(T_s - T_\infty)}$$

∴ The minimum h required exists when T_s is at maximum:

$$h_{min} = \frac{9W}{25\times10^{-6}m^2 \cdot (80°C - 20°C)} = \boxed{6000\ W/m^2 \cdot K}$$

TABLE 1.1 Typical values of the convection heat transfer coefficient

Process	h ($W/m^2 \cdot K$)
Free convection	
Gases	2–25
Liquids	50–1000
Forced convection	
Gases	25–250
Liquids	100–20,000
Convection with phase change	
Boiling or condensation	2500–100,000

From the textbook, it looks like forced convection with liquid and convection with boiling or condensation can reach the $h > 6000\ W/m^2 \cdot K$.

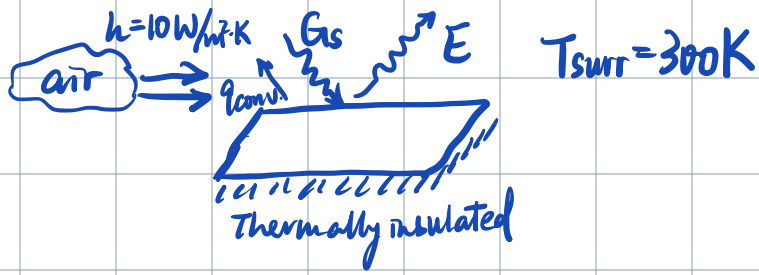
Problem 3.

Known: $G_s = 750\,W/m^2$, $T_\infty = T_{sur} = 300K$, $h_{air} = 10\,W/m^2\cdot K$

Find: (a) $T_{roof} = ?$ (b) dominant heat transfer mode

Property: roof's material — $\varepsilon = 0.18$, $\alpha_s = 0.76$

Schematic:



Assumption: (1) Bottom surface is thermally insulated. (2) roof is a grey body with $\varepsilon = 0.18$

Analysis: (a) As the energy absorbed should equal to the energy loss by emission and the energy loss by convection:

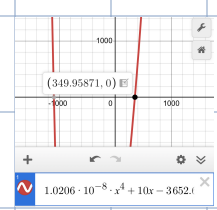
$$\alpha_s G = \varepsilon\sigma(T_s^4 - T_{sur}^4) + h(T_s - T_\infty)$$

Plug in: $0.76 \times 750\,W/m^2 = 0.18 \times 5.67\times10^{-8}\,W/m^2K^4 \cdot (T_s^4 - 300K^4) + 10\,W/m^2K\,(T_s - 300K)$

$$570 = 1.0206\times10^{-8}\,T_s^4 - 82.6686 + 10T_s - 3000$$

$$1.0206\times10^{-8}\,T_s^4 + 10T_s = 3652.6686$$

By plotting, we get $\boxed{T_s = 349.96K}$ or $\underline{T_s = -1093.39K}$ reject.



$T_s = 349.96K = 76.81°C$ which could be a bit hot but not to an extent that would burn or hurt the cat's feet, so she should be comfortable walking on it.

(b) As we can see from the calculation above with $T_s = 350K$:

Radiation: $q''_{rad} = \varepsilon\sigma(T_s^4 - T_{sur}^4)$

$= 0.18 \times 5.67 \times 10^{-8} W/m^2K^4 (349.96^4 - 300K^4)$

$= 70.42 W$

Convection: $q''_{conv} = h(T_s - T_\infty)$

$= 10 W/m^2K (349.96K - 300K)$

$= 499.6 W$

Therefore, the outgoing heat flux by convection is more dominant than that by radiation.

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #1

Posted: Monday, Jan 26 **Due: Tuesday, Feb 3, 11:59 pm**

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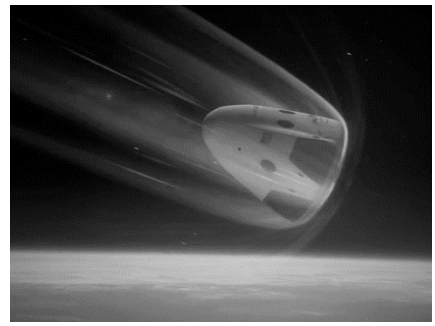
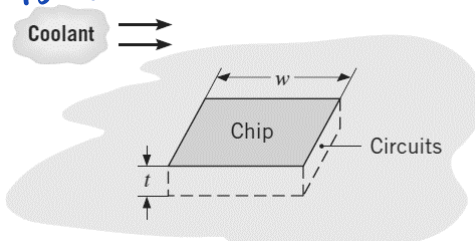
1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.



Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.

$T_\infty = 20°C$



(a) What is the steady-state temperature difference between back and front surfaces of the chip?

(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

Fourier's law

$$q_x = -kA\frac{dT}{dx}$$

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

(a) Consider a day of solar irradiation $G_s = 750$ W/m^2, ambient temperature $T_\infty = 300$ K, surrounding temperature $T_{surr} = 300$ K, and heat transfer coefficient of air flow $h = 10$ W/(m$^2\cdot$ K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

compare outgoing heat flux from surface

(i.e. heat transfer out of roof)